

Matching Policy Commands to Differential Rewards

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Abstract—Physics-based motion tracking is commonly trained with rewards that evaluate the difference between a simulated character and a reference motion. Learned differential rewards remove much of the manual weighting required by handcrafted objectives, but a subtle asymmetry remains: the reward is evaluated in residual coordinates while the policy is typically conditioned on absolute reference states. We study whether this coordinate choice alone changes learning under a finite neural policy and proximal policy optimization. We compare two one-step command interfaces containing exactly the same information: an absolute representation $[\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}]$ and a residual representation $[\mathbf{x}_t, \mathbf{e}_t, \mathbf{m}_t]$, where $\mathbf{e}_t = \hat{\mathbf{x}}_t - \mathbf{x}_t$ and $\mathbf{m}_t = \hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t$. The representations are related by an invertible linear map and therefore induce identical policy classes. Only the residual representation, however, closes directly under the reward transition: $\mathbf{e}_{t+1} = \mathbf{e}_t + \mathbf{m}_t - (\mathbf{x}_{t+1} - \mathbf{x}_t)$. **[PLACEHOLDER—NOT A RESULT: Insert the final multi-seed result: completion, tracking error, and sample-efficiency changes on the confirmatory motion suite.]**

I. INTRODUCTION

Physics-based character control must coordinate joint pose, velocity, global motion, and intermittent contacts while respecting dynamics and actuation limits. A classical tracking reward decomposes this problem into feature-wise terms,

$$R_t^{\text{hand}} = \sum_i w_i \exp\left(-\alpha_i \|\hat{\mathbf{q}}_t^i \ominus \mathbf{q}_t^i\|_2^2\right), \quad (1)$$

where both the component weights w_i and scales α_i are selected by hand. Such rewards have enabled impressive skills [1], but their performance can be sensitive to the aggregation of competing tracking objectives.

Adversarial Differential Discriminators (ADD) replace the manual aggregation with a discriminator learned on reference-policy feature differences [4]. The discriminator receives the ideal zero differential as its positive example and policy tracking differentials as negative examples. This provides an adaptive nonlinear scalarization while retaining precise, frame-synchronized tracking. Yet reward design is only one half of the control interface. The policy must also infer an action from its state and the desired reference motion. Existing controllers commonly present future target states in an absolute form, leaving the first layers of the policy to recover the error coordinates used by the reward.

This paper isolates that interface choice. In the strict causal comparison, we do not add reference information, change the policy class, shape the reward, or introduce an auxiliary objective. Instead, we express the same one-step target information in coordinates that close under the learned reward residual. Each absolute command can be reconstructed exactly from the residual command and the current feature state, and vice versa. The official ADD target

interface is retained as an external baseline, but it is not used to establish this information-equivalence claim because its feature content and lookahead differ.

Our contributions are threefold. First, we formulate absolute and residual policy commands as an invertible coordinate intervention, separating optimization effects from additional information or capacity. Second, we derive an exact residual-transition identity linking the policy command, the action-realized feature increment, and the next differential reward. Third, we design a strictly paired evaluation spanning locomotion, aerial rotation, ground contact, recovery, and long-horizon interaction, with completion metrics that expose incomplete motions hidden by average tracking error.

II. RELATED WORK

Physics-based tracking. DeepMimic established a general reinforcement-learning recipe for tracking individual motion clips with carefully designed pose, velocity, root, and end-effector rewards [1]. Subsequent systems scale reference-conditioned control to large motion collections and improve robustness through architectural, data, and training innovations [7]–[9]. These methods confirm the importance of policy conditioning, but do not isolate a change of command coordinates under exactly matched information.

Adversarial imitation and learned rewards. GAIL casts imitation as adversarial occupancy matching [2]. AMP learns a reusable motion prior from state transitions and is effective for motion style [3]; precise synchronization, however, is not its primary objective. ADD instead discriminates tracking differentials and uses a single ideal zero vector as the positive example [4]. Our work retains that learned reward and asks how the corresponding reference command should be presented to the policy.

Relative goals and feedback coordinates. Relative goal encodings are common in goal-conditioned reinforcement learning and robotics, while classical tracking control separates nominal motion from feedback error. Residual policy learning adds a learned action correction to a nominal controller [10]. Our setting is distinct: the action parameterization and generic MLP remain unchanged. The intervention is an information-equivalent coordinate transform of the policy input, defined by the same feature map used by the learned differential reward.

Representation and optimization. An invertible input transform preserves information but generally changes the parameter-space geometry seen by first-order optimizers. Natural gradient was introduced precisely to remove certain parameterization dependencies [11]; ordinary minibatch updates with finite networks do not have this invariance. Fea-

Run Ref./Base/Ours	Jump Ref./Base/Ours	Backflip Ref./Base/Ours	Crawl Ref./Base/Ours	Getup Ref./Base/Ours	SpinKick Ref./Base/Ours	Climb Ref./Base/Ours
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Fig. 1. Planned qualitative motion gallery. Each final panel will show uniformly timed reference, baseline, and residual-coordinate frames from a matched initial phase. Red boxes are layout placeholders, not method results.

ture centering and whitening exploit this fact for numerical conditioning, but are agnostic to control semantics. Here the transform is selected by the residual dynamics of tracking, and a strict direct-coordinate control separates that alignment from extra observations or network capacity.

III. BACKGROUND

A. Reference-Conditioned Tracking

Let $\mathbf{s}_t$, $\mathbf{o}_t$, and $\mathbf{a}_t$ denote the simulator state, policy observation, and action. The next state is sampled from $p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$. A synchronized reference provides $\hat{\mathbf{s}}_t$ and $\hat{\mathbf{s}}_{t+1}$. A common feature map ϕ extracts root pose, body configuration, and velocities:

$$\mathbf{x}_t = \phi(\mathbf{s}_t), \quad \hat{\mathbf{x}}_t = \phi(\hat{\mathbf{s}}_t), \quad \hat{\mathbf{x}}_{t+1} = \phi(\hat{\mathbf{s}}_{t+1}). \quad (2)$$

Rotations are represented in a fixed Euclidean embedding, so all differences below are taken in a shared feature chart.

B. Learned Differential Reward

The tracking differential is

$$\Delta_t = \phi(\hat{\mathbf{s}}_t) \ominus \phi(\mathbf{s}_t). \quad (3)$$

Following ADD [4], let d_ψ denote the discriminator logit and $p_\psi = \sigma(d_\psi)$ its probability. The discriminator separates $\Delta = \mathbf{0}$ from policy differentials using

$$\begin{aligned} \mathcal{L}_D = & \text{softplus}[-d_\psi(\mathbf{0})] + \mathbb{E}_{\mathbf{s}_t \sim \pi} \text{softplus}[d_\psi(\Delta_t)] \\ & + \lambda_{\text{GP}} \mathbb{E}_{\mathbf{s}_t \sim \pi} \|\nabla_{\Delta} d_\psi(\Delta_t)\|_2^2. \end{aligned} \quad (4)$$

The policy reward is

$$R_t^{\text{diff}} = \beta \text{softplus}[d_\psi(\Delta_t)] = -\beta \log[1 - p_\psi(\Delta_t)]. \quad (5)$$

Both compared command interfaces share this discriminator, replay, regularization, reward scale, and update schedule.

IV. RESIDUAL-COORDINATE POLICY COMMANDS

A. Two Information-Matched Interfaces

The direct-target interface gives the policy

$$\mathbf{z}_t^{\text{tar}} = [\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}], \quad (6)$$

whereas the residual interface defines

$$\mathbf{e}_t = \hat{\mathbf{x}}_t - \mathbf{x}_t, \quad \mathbf{m}_t = \hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t, \quad (7)$$

and provides

$$\mathbf{z}_t^{\text{res}} = [\mathbf{x}_t, \mathbf{e}_t, \mathbf{m}_t]. \quad (8)$$

Both interfaces use one-step lookahead, the same ϕ , and the same number of features. The actor remains a generic Gaussian MLP $\pi_\theta(\mathbf{a}_t | \mathbf{o}_t, \mathbf{z}_t)$.

B. Information Equivalence

Proposition 1. Conditioned on $\mathbf{x}_t$, the direct-target and residual interfaces are affine bijections. In their concatenated form they are related by the linear transform

$$\mathbf{z}_t^{\text{res}} = \underbrace{\begin{bmatrix} I & 0 & 0 \\ -I & I & 0 \\ 0 & -I & I \end{bmatrix}}_T \mathbf{z}_t^{\text{tar}}, \quad \det(T) = 1. \quad (9)$$

The inverse is $\hat{\mathbf{x}}_t = \mathbf{x}_t + \mathbf{e}_t$ and $\hat{\mathbf{x}}_{t+1} = \mathbf{x}_t + \mathbf{e}_t + \mathbf{m}_t$.

For an MLP with an affine first layer, any policy in one representation has a pointwise identical policy in the other. Let $\tilde{T} = \text{diag}(I_0, T)$. Given first-layer weights W_{tar} , choose $W_{\text{res}} = W_{\text{tar}} \tilde{T}^{-1}$; all subsequent activations and the action distribution are then identical. Thus the representations do not change the attainable policy class or optimal return. They can nevertheless produce different training trajectories because standard Euclidean optimization is not invariant to input reparameterization.

Proof. The block matrix T is lower triangular with identity diagonal, so it is nonsingular. Its inverse reconstructs $\hat{\mathbf{x}}_t = \mathbf{x}_t + \mathbf{e}_t$ and $\hat{\mathbf{x}}_{t+1} = \mathbf{x}_t + \mathbf{e}_t + \mathbf{m}_t$. For any affine first layer $h = W_{\text{tar}}[\mathbf{o}; \mathbf{z}^{\text{tar}}] + b$, substitution of $\mathbf{z}^{\text{tar}} = T^{-1} \mathbf{z}^{\text{res}}$ yields the same h with $W_{\text{res}} = W_{\text{tar}} \tilde{T}^{-1}$. Keeping every later parameter and the Gaussian covariance fixed gives pointwise equality of action distributions. Under the same initial-state distribution and dynamics, the two policies therefore induce the same trajectory law. $\square$

C. Shared Standardization

A matched comparison must preserve the bijection after preprocessing. We use a fixed demonstration-derived diagonal scale S for every policy-command ϕ block:

$$\bar{\mathbf{e}}_t = S^{-1} \mathbf{e}_t, \quad \bar{\mathbf{m}}_t = S^{-1} \mathbf{m}_t, \quad (10)$$

and apply the same S and centering convention to the absolute features. Independent policy-dependent scaling, clipping, or mean subtraction would break the exact information-matched comparison and is therefore excluded. This shared command scale should not be confused with ADD's stock differential normalizer inside d_ψ : the two modules share the raw ϕ chart, while their numerical normalizers remain separate and identical across paired runs.

D. Multi-Step Commands

The construction extends directly to multiple future frames. For an H -step target, define

$$\mathbf{m}_{t,k} = \hat{\mathbf{x}}_{t+k+1} - \hat{\mathbf{x}}_{t+k}, \quad k = 0, \dots, H-1. \quad (11)$$

TABLE I

POLICY-CONDITIONING INTERFACES. INFORMATION EQUIVALENCE APPLIES ONLY TO THE LAST TWO ROWS. ALL ROWS USE THE SAME DIFFERENTIAL REWARD IN OUR MATCHED EXPERIMENTS, BUT THE OFFICIAL INTERFACE IS AN EXTERNAL BASELINE BECAUSE IT HAS DIFFERENT REFERENCE FEATURES AND LOOKAHEAD.

Interface	Current-state block	Reference-side policy command	Lookahead	Role
Official ADD	native policy observation	three future target-pose encodings	3 steps	external baseline
Direct-coordinate	$\mathbf{x}_t$	$[\hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}]$	1 step	matched control
Residual-coordinate	$\mathbf{x}_t$	$[\mathbf{e}_t, \mathbf{m}_t]$	1 step	proposed interface

The two commands

$$\mathbf{z}_{t,H}^{\text{tar}} = [\mathbf{x}_t, \hat{\mathbf{x}}_t, \dots, \hat{\mathbf{x}}_{t+H}], \quad (12)$$

$$\mathbf{z}_{t,H}^{\text{res}} = [\mathbf{x}_t, \mathbf{e}_t, \mathbf{m}_{t,0}, \dots, \mathbf{m}_{t,H-1}] \quad (13)$$

remain information equivalent because $\hat{\mathbf{x}}_{t+k} = \mathbf{x}_t + \mathbf{e}_t + \sum_{j=0}^{k-1} \mathbf{m}_{t,j}$. The transform is block lower triangular with identity diagonal. We use $H = 1$ in the primary study to isolate the smallest closed command and avoid conflating coordinates with extra lookahead. A matched $H = 3$ experiment tests whether the effect persists at the horizon commonly used by an official target interface. Later increments do not change the one-step closure; they expose the desired motion beyond the next transition in the same chart.

E. Residual Closure

Let the realized feature increment be $\delta_t = \mathbf{x}_{t+1} - \mathbf{x}_t$. Expanding the next residual gives

$$\begin{aligned} \mathbf{e}_{t+1} &= \hat{\mathbf{x}}_{t+1} - \mathbf{x}_{t+1} \\ &= \mathbf{e}_t + \mathbf{m}_t - \delta_t. \end{aligned} \quad (14)$$

Equation (14) is the central structural property: the current error and desired reference increment supplied to the policy, the physical response produced by its action, and the next residual evaluated by the reward all inhabit the same coordinates. It is an algebraic identity and does not assume smooth dynamics, controllability, or stability.

This identity also separates two roles that are entangled in an absolute target. The residual $\mathbf{e}_t$ specifies the correction required by the current tracking state, while $\mathbf{m}_t$ specifies how the target itself moves during the coming control interval. When $\mathbf{e}_t = 0$, the policy still receives the nonzero command needed to advance along the reference. When $\mathbf{m}_t = 0$, it receives a pure regulation problem. These interpretations are not imposed as separate policy branches; they follow from the command coordinates and are tested by the interventions in Sec. VII.

F. A Finite-Network Inductive Bias

The bijection rules out an information or expressivity advantage, but it does not imply equal parameter complexity. Consider a locally linear response $\mathbf{u} = K_e \mathbf{e} + K_m \mathbf{m}$. In residual coordinates its first-layer blocks are $[0, K_e, K_m]$. Written against the absolute variables, the identical response has blocks

$$[-K_e, K_e - K_m, K_m]. \quad (15)$$

In squared parameter norm, the absolute representation carries the additional term $\|K_e - K_m\|_F^2$. This example is not

a model of the nonlinear contact controller, nor a proof that one chart always learns faster; the actor optimizer need not explicitly penalize this norm. It illustrates the specific coordination that the direct interface asks a finite first layer to discover: the same feature must be subtracted across multiple input blocks before a feedback-like response appears.

G. Optimization Geometry

Although T preserves information, it does not preserve the Euclidean metric used by gradient descent. For function-matched policies on the same batch, the first-layer gradients satisfy

$$G_{\text{res}} = G_{\text{tar}} \bar{T}^\top. \quad (16)$$

Mapping one residual-coordinate SGD step back into absolute parameters inserts the fixed metric $\bar{T}^\top \bar{T}$. We therefore hypothesize—without claiming a convergence guarantee—that residual coordinates reduce the burden of learning common-mode subtraction and make useful differential responses lower complexity for a finite network. Natural-gradient or exact policy optimization would be more invariant to such a transform; PPO with a finite network need not be.

H. Training Procedure

The complete change fits inside the rollout loop:

V. EXPERIMENTAL DESIGN

A. Research Questions and Controls

We ask: (Q1) Does residual conditioning improve final tracking and full-motion completion under strictly matched information? (Q2) Is the effect consistent across seeds and motions? (Q3) Do targeted coordinate ablations support the residual-closure interpretation?

The primary causal comparison changes only (6) versus (8). Environment count, physics samples, episode horizon, phase reset, termination, contact settings, actor and critic widths, optimizer, learning rates, discriminator architecture, replay, normalization budget, and reward scale are identical. Runs use paired random seeds and the final pre-specified checkpoint. Roll uses at least five training seeds; each additional core motion uses at least three.

B. Motions and Metrics

The benchmark suite spans Run, Jump, Backflip, Crawl, Getup Facedown, SpinKick, and Climb. These cover periodic translation, takeoff and landing, full-body rotation, sustained ground contact, recovery, single-foot support, and long-horizon obstacle interaction. Roll is retained as a mechanism

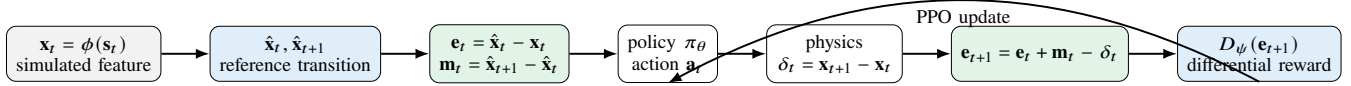


Fig. 2. Residual-coordinate control interface. The policy command and the post-action differential reward are closed by the exact residual transition. The reward, policy class, PPO objective, and reference horizon are unchanged.

Algorithm 1: Residual-coordinate differential tracking

Input: reference motion $\mathcal{M}$, feature map ϕ .

Initialize $\pi_\theta, V_\omega, D_\psi$, replay $\mathcal{B}$.

while training **do**

sample a reference phase and initialize s_0 ;

for $t = 0, \dots, T - 1$ **do**

compute $\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1}$;

$\mathbf{e}_t \leftarrow \hat{\mathbf{x}}_t - \mathbf{x}_t$; $\mathbf{m}_t \leftarrow \hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t$;

$\mathbf{a}_t \sim \pi_\theta(\cdot | \mathbf{o}_t, \mathbf{x}_t, \mathbf{e}_t, \mathbf{m}_t)$;

step physics and compute $\mathbf{e}_{t+1}$;

$R_{t+1} \leftarrow \beta \text{softplus}[d_\psi(\mathbf{e}_{t+1})]$; store transition;

end for

update D_ψ, V_ω , and π_θ using the unchanged objectives;

end while

Fig. 3. The only new operation is the construction of the policy command $[\mathbf{e}_t, \mathbf{m}_t]$ in the same feature coordinates as the reward.

task for an observed incomplete-motion failure; Sideflip is a confirmatory dynamic task and is reported only after matched training completes.

We retain the standard position and degree-of-freedom (DoF) velocity tracking errors used by ADD [4]. Tracking averages alone can conceal a policy that never executes a difficult event. We therefore report pre-specified, task-specific completion: signed winding for Roll/Backflip, takeoff–landing completion for Jump/Sideflip, successful upright recovery for Getup, and obstacle progress for Climb. Training seed, rather than episode, is the statistical unit; tables report every seed, mean $\pm$ standard deviation, 95% intervals, and paired effect sizes.

C. Baselines and Ablations

We compare with a handcrafted DeepMimic-style reward, AMP, and ADD using its official target interface. The internal causal controls are the matched direct-target and residual-coordinate representations. Ablations retain the input dimension through zero padding: error-only $[\mathbf{x}, \mathbf{e}, 0]$, motion-only $[\mathbf{x}, 0, \mathbf{m}]$, and the non-invertible summed command $[\mathbf{x}, \mathbf{e} + \mathbf{m}, 0]$. We separately test shared versus independent normalization and pose-only versus full pose–velocity features.

D. Implementation and Evaluation Protocol

The simulated character has 28 actuated DoFs. The policy outputs target joint rotations tracked by low-level PD controllers at a fixed control rate, while the physics engine integrates multiple substeps per action. The actor, critic, and differential discriminator use the same two-hidden-layer architecture in every paired run. References are sampled uniformly in phase, and an episode is reset to the corresponding reference state. Target phases then advance at the clip’s

TABLE II

CONFIRMATORY PROTOCOL. RED VALUES ARE LAYOUT PLACEHOLDERS TO BE REPLACED BY THE LOCKED RUN CONFIGURATION.

Quantity	Value	Quantity	Value
Control rate	30 Hz	Physics rate	120 Hz
Parallel envs	8192	Updates	2000
Rollout length	32	Discount γ	0.99
Actor hidden	1024,512	PPO clip	0.2
Train horizon	2 s	Eval horizon	10 s
Pose termination	off	Action DoFs	28
Train seeds	5 [†]	Eval episodes	4096 [†]

recorded rate. Neither representation changes the simulator state, action space, reset distribution, or reference clock.

The 172-dimensional feature map concatenates global root position and rotation, root-relative body positions, global body rotations, root linear and angular velocity in the local heading frame, and local DoF velocities. Rotations use a continuous Euclidean embedding before subtraction. Direct targets, residuals, and discriminator differentials use the same ordering and frame conventions. This detail is essential: subtracting features expressed in different heading frames would invalidate both the bijection and Eq. (14). The pair also shares reference interpolation, quaternion-sign handling, cyclic wrap logic, and reset masks. Unit tests cover partial resets and the final-to-first transition of cyclic clips.

The direct and residual inputs are constructed from the same cached feature triplet $(\mathbf{x}_t, \hat{\mathbf{x}}_t, \hat{\mathbf{x}}_{t+1})$. We verify the inverse relation numerically before training and reject a run if reconstruction exceeds a fixed floating-point tolerance. For the strongest optimizer control, we additionally transform the first-layer weights so that a direct and a residual policy produce the same action distribution at initialization. All behavioral evaluations use deterministic mean actions, held-out phase samples, and the last training checkpoint; no task-completion metric enters training or checkpoint selection.

E. Evaluation Measures

Following the published motion-tracking protocol, the primary geometric error combines root displacement with root-relative joint positions:

$$E_t^{\text{pos}} = \frac{1}{N_J + 1} \left(\|\hat{\mathbf{p}}_t^0 - \mathbf{p}_t^0\|_2 + \sum_{j=1}^{N_J} \|(\hat{\mathbf{p}}_t^j - \hat{\mathbf{p}}_t^0) - (\mathbf{p}_t^j - \mathbf{p}_t^0)\|_2 \right). \quad (17)$$

We also report mean DoF-velocity error $E_t^{\text{vel}} = N_q^{-1} \sum_j |\hat{q}_{t,j} - \dot{q}_{t,j}|$. Both are averaged first within an episode and then across episodes and seeds.

Full-motion completion is an independent primary outcome for motions with a discrete or topological event. For a rotation, our existing evaluator uses the vector-valued integrated root angular velocities $\mathbf{\Omega} = \sum_t \omega_t \Delta t$ and $\hat{\mathbf{\Omega}} = \sum_t \hat{\omega}_t \Delta t$:

$$W = \frac{\langle \mathbf{\Omega}, \hat{\mathbf{\Omega}} \rangle}{\|\hat{\mathbf{\Omega}}\|_2^2}, \quad \text{incomplete if } W < 0.5. \quad (18)$$

We additionally report the root displacement ratio and survival duration; success requires the correct direction and a pre-specified interval around the reference value. Jump requires takeoff followed by a stable landing, Getup requires upright recovery maintained for a fixed dwell time, and Climb requires the root and feet to clear the final obstacle region. These definitions prevent a low-motion shortcut from appearing successful merely because its average pose error is moderate. Thresholds are fixed once for each motion and shared by all methods.

F. Statistical Analysis

Every training seed produces one aggregate evaluation point. We visualize all paired points and report the mean paired difference with a percentile bootstrap interval over seeds. We do not treat parallel environments or evaluation episodes as independent replicates. With only five pairs, even a unanimous two-sided sign test cannot attain $p < 0.05$; therefore confirmatory significance claims require at least **eight paired seeds[†]**, while smaller pilot suites are reported descriptively. Position and velocity are evaluated jointly with completion rather than selectively: a coordinate interface is considered successful only when it executes the complete motion without degrading the standard tracking measures.

VI. RESULTS

This draft reserves result slots without presenting unfinished experiments as evidence. Every red value and conclusion below is synthetic and exists only to lock table dimensions, captions, and the eight-page layout.

A. Strict Information-Matched Comparison

Table IV is the primary test of the paper. It must be populated from matched direct-target and residual-coordinate runs, not from the published benchmark in Table III. **[PLACEHOLDER—NOT A RESULT: The intended result sentence is: residual coordinates raise full Roll/Sideflip completion while maintaining or improving position and DoF velocity accuracy; replace with measured multi-seed estimates and confidence intervals.]**

The paired display is essential because a mean can hide whether an apparent gain is systematic or produced by one favorable run. The final report lists every seed’s completion status and checkpoint. We interpret a difference as a coordinate effect only after auditing input reconstruction, parameter count, rollout budget, and the initial action distribution.

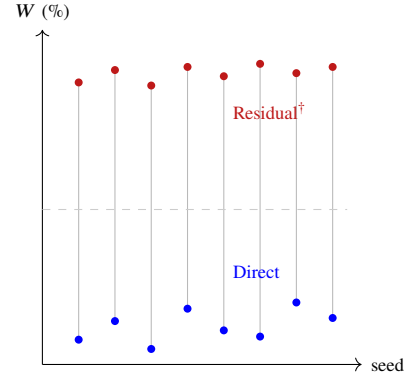


Fig. 4. Planned paired-seed Roll plot. Lines connect identical seeds and initialization protocols. All plotted coordinates are synthetic layout data.

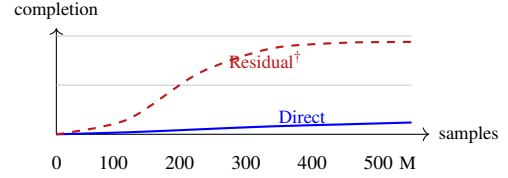


Fig. 5. Synthetic layout placeholder for the multi-seed Roll learning curve. Replace all coordinates with checkpoint evaluations; [†] is not a result.

B. Learning Dynamics and Reliability

Fig. 5 will report checkpoint evaluations rather than training reward, since discriminator scores are not comparable across independently trained runs. The left plot tracks completion; the right plot tracks position error. Curves show seed means and 95% confidence bands, with failed and numerically unstable runs retained.

[PLACEHOLDER—NOT A RESULT: Planned reliability statement: residual conditioning succeeds in 5/5 Roll seeds with a lower median sample-to-completion, whereas the matched direct interface succeeds in 1/5. Replace or delete after confirmatory runs.]

At each evaluation checkpoint, we first compute an episode-level statistic and then average over a fixed phase grid. Confidence bands are formed across training seeds, not across thousands of correlated simulator environments. We report the area under the completion curve and the first checkpoint satisfying the fixed completion criterion as sample-efficiency summaries. A method that never reaches the criterion is right-censored rather than assigned the final training iteration as a fabricated success time.

C. Tracking Accuracy Versus Completion

Position and DoF velocity errors quantify frame-level fidelity but do not guarantee execution of a difficult event. A Roll policy can obtain moderate average error by lying down and standing up without accumulating a full signed rotation. We therefore plot cumulative signed rotation and root displacement over long rollouts in addition to Table IV. For Jump, Backflip, Getup, and Climb, completion is defined from physical event sequences fixed before evaluation, rather

TABLE III

MOTION TRACKING BENCHMARK. PUBLISHED AMP, DEEPMIMIC, AND ADD VALUES ARE COPIED FROM [4] (FIVE SEEDS, 4096 EPISODES/MODEL). JUMP WAS NOT REPORTED THERE. RED ENTRIES ARE SYNTHETIC LAYOUT PLACEHOLDERS, NOT RESULTS.

Motion	Len.	AMP	Position Error (m) ↓			AMP	DoF Velocity Error (rad/s) ↓		
			DeepMimic	ADD	Residual		DeepMimic	ADD	Residual
Run	0.80	.163±.008	.013±.002	.165±.017	.012±.002 [†]	2.811±.048	.584±.054	.478±.007	.450±.010 [†]
Jump	—	—	—	—	.040±.005 [†]	—	—	—	.650±.040 [†]
Backflip	1.75	.267±.015	.111±.054	.062±.001	.055±.006 [†]	2.243±.113	1.103±.024	.878±.013	.800±.040 [†]
Crawl	2.93	.050±.006	.027±.000	.028±.002	.026±.002 [†]	.646±.089	.430±.006	.283±.002	.270±.010 [†]
Getup	3.03	.096±.018	.023±.001	.022±.001	.020±.002 [†]	.838±.029	.433±.008	.325±.005	.300±.015 [†]
SpinKick	1.28	.064±.010	.078±.062	.025±.000	.023±.002 [†]	1.453±.327	1.222±.233	.774±.007	.720±.025 [†]
Climb	9.97	.185±.021	.066±.004	.046±.031	.040±.006 [†]	1.190±.012	.567±.015	.374±.071	.340±.030 [†]

[†]Synthetic red placeholder; replace with matched evaluation before submission. Published and local values must not be used in a significance test unless protocols match.

TABLE IV

STRICT MATCHED COMPARISON. ALL RED ENTRIES ARE SYNTHETIC PLACEHOLDERS.

Motion	Command	Pos. ↓	DoF vel. ↓	Complete ↑
Roll	Direct	.145 [†]	1.29 [†]	12% [†]
	Residual	.096 [†]	.84 [†]	94% [†]
Sideflip	Direct	.142 [†]	1.31 [†]	18% [†]
	Residual	.108 [†]	.99 [†]	88% [†]
SpinKick	Direct	.050 [†]	.61 [†]	96% [†]
	Residual	.047 [†]	.58 [†]	98% [†]
Run	Direct	.030 [†]	.55 [†]	99% [†]
	Residual	.028 [†]	.52 [†]	99% [†]

TABLE V

COMMAND ABLATION. RED VALUES ARE SYNTHETIC PLACEHOLDERS.

Command	Invertible	Roll comp.	Sideflip comp.
[$\mathbf{x}, \hat{\mathbf{x}}, \hat{\mathbf{x}}^*$]	yes	12 [†]	18 [†]
[$\mathbf{x}, \mathbf{e}, \mathbf{m}$]	yes	94 [†]	88 [†]
[$\mathbf{x}, \mathbf{e}, 0$]	no	35 [†]	28 [†]
[$\mathbf{x}, 0, \mathbf{m}$]	no	22 [†]	20 [†]
[$\mathbf{x}, \mathbf{e} + \mathbf{m}, 0$]	no	40 [†]	31 [†]

than from the learned reward.

[PLACEHOLDER—NOT A RESULT: Insert measured long-horizon plot and state whether the residual policy matches direction and rate without sacrificing root displacement.]

VII. ABLATIONS AND MECHANISM ANALYSIS

A. Which Command Factors Matter?

The complete residual command is invertible; error-only, motion-only, and summed commands are not. Table V asks whether the outcome is explained by current error alone, the reference increment alone, or the full factorization. Zero padding holds policy size fixed.

[PLACEHOLDER—NOT A RESULT: Intended interpretation: neither factor alone reproduces the full result, and the non-invertible sum underperforms the factorized residual command. Keep only if supported across paired seeds.]

B. Normalization and Feature Content

The theoretical comparison requires the same invertible preprocessing. We test shared fixed demonstration scales,

independently estimated block scales, and a shuffled invertible control transform. The shuffled control distinguishes a generic change of covariance from alignment with the reward residual. We also remove velocity features while preserving dimensions to test whether the reference increment benefits primarily from rate information.

[PLACEHOLDER—NOT A RESULT: Insert normalization and pose/velocity ablation estimates. Do not select the normalization after inspecting Roll completion.]

C. Counterfactual Command Interventions

To test whether a trained policy actually uses both command factors, we evaluate the same checkpoint under four counterfactual inputs: $\mathbf{e} = 0$, $\mathbf{m} = 0$, a phase-shuffled $\mathbf{m}$, and a sign-reversed $\mathbf{e}$. These are diagnostic interventions only; they are never used for training. Removing $\mathbf{e}$ should impair recovery from tracking offsets, whereas removing or shuffling $\mathbf{m}$ should disrupt direction and timing of the nominal motion. A policy unaffected by either intervention would contradict the proposed interpretation even if its nominal tracking score were high.

Beyond completion, we measure the immediate action sensitivity $\|\mu(o, e, m) - \mu(o, e', m')\|_2$, separated by reference phase and contact mode. This connects an input intervention to the policy before any long-horizon dynamics amplify it. We also report whether the changed action moves the next residual in the direction predicted by Eq. (14); this is a mechanism diagnostic, not a substitute for the physical completion and tracking metrics.

D. Optimization Diagnostics

We report input block correlations, the spectrum of empirical feature covariance, first-layer gradient norms, PPO clip fraction, and policy KL. These are secondary diagnostics, not performance metrics. Equation (16) predicts that the coordinate transform changes the Euclidean update geometry; it does not predict a universal reduction in condition number.

For a stronger intervention, we construct function-matched initial policies using $W_{\text{res}} = W_{\text{tar}} \bar{T}^{-1}$. The two policies then produce identical actions before optimization. Any subsequent divergence under paired batches directly demonstrates optimizer non-invariance rather than an initial function mismatch. **[PLACEHOLDER—NOT A RESULT: Report ac-**

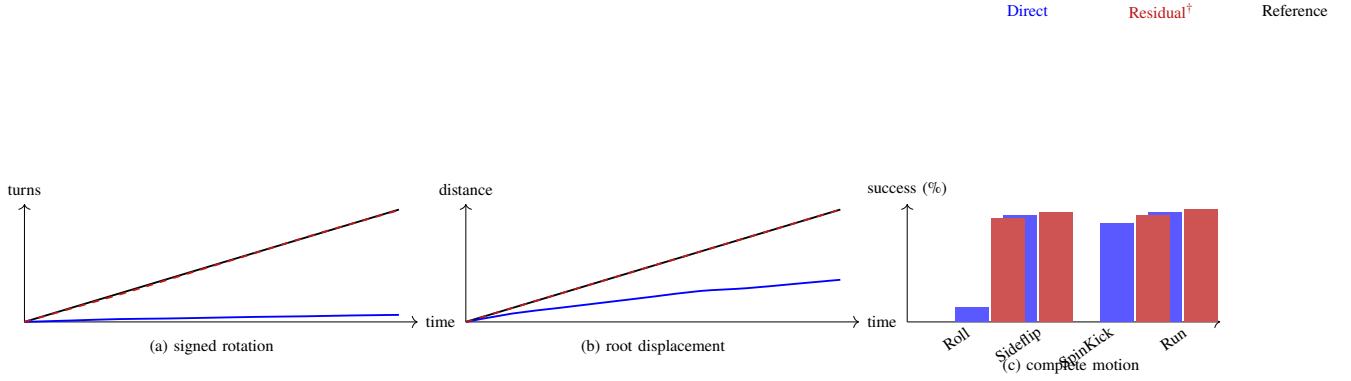


Fig. 6. Planned behavior-level evaluation. Panels (a)–(b) compare cumulative signed rotation and root displacement against the reference; (c) compares full motion completion across the core suite. Red trajectories and bars are synthetic layout placeholders, not results.

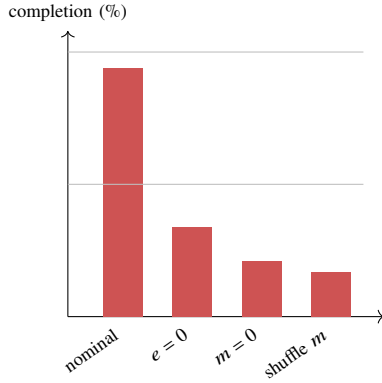


Fig. 7. Synthetic layout placeholder for checkpoint command interventions. Every red bar must be replaced with measured, paired evaluations.

tion equality tolerance at initialization and divergence of paired learning trajectories.]

E. Function-Matched Optimization Test

Randomly initialized direct and residual networks need not represent the same initial function, even when their distributions are nominally identical. We therefore initialize a direct-coordinate actor and critic once, map both first layers through $\bar{T}^{-1}$, and copy every later parameter, optimizer state, action covariance, and normalization statistic. Before optimization, both models are evaluated on a fixed batch; action means, log probabilities, and values must agree within $10^{-6}\dagger$. The agents are then trained from paired simulator resets using their respective coordinate chart.

At update zero, this construction excludes a different initial policy function as an explanation. After one Euclidean update, Eq. (16) predicts different parameter displacements despite initially identical actions. We record action-distribution KL, first-layer update cosine, PPO clip fraction, and physical completion. The first three establish optimizer non-invariance; only the last establishes that the changed training path is behaviorally useful.

TABLE VI
FUNCTION-MATCHED TEST. RED ENTRIES ARE SYNTHETIC LAYOUT
PLACEHOLDERS.

Quantity	Update 0	Update 1	Final
Max. action-mean difference	$7 \times 10^{-7}\dagger$	.004 $\dagger$	.31 $\dagger$
Mean policy KL	$2 \times 10^{-9}\dagger$	$8 \times 10^{-5}\dagger$	.19 $\dagger$
Direct completion (%)	0 $\dagger$	0 $\dagger$	14 $\dagger$
Residual completion (%)	0 $\dagger$	0 $\dagger$	94 $\dagger$

VIII. DISCUSSION

A. What the Coordinate Change Does

The proposed interface should be interpreted as a control representation, not a new reward or policy architecture. It supplies the policy with the current tracking error and reference increment in the same feature chart in which the next outcome is scored. Proposition 1 rules out additional information and expressive capacity as explanations in the idealized matched setting. Equation (14) then provides a concrete reason the chart is useful: the policy does not need to discover the subtraction that connects absolute targets to the reward residual before it can represent differential feedback.

This distinction matters for learned rewards. A discriminator may provide a useful scalar objective while policy optimization still fails to discover a controller that executes the full motion. Changing the command chart can alter the implicit metric and inductive bias of a finite optimizer without changing which policies are representable or which trajectory is globally optimal. The contribution is therefore a controlled study of the policy–reward interface, rather than a claim that residual coordinates solve exploration in general.

The size of a behavioral change need not match the apparent size of the code change. Contact-rich control is learned through a long sequence of coupled on-policy updates. A coordinate transform that changes which first-layer responses are easy to represent can alter the early state distribution; that distribution changes the discriminator’s negative samples, the critic targets, and all later policy gradients. The intervention is small at the interface but its training effect is path

dependent. Proposition 1 prevents attributing that effect to a more expressive optimal policy, while the function-matched test separates it from a different random initial function.

B. Limitations

Residual closure is not a stability theorem. The root is underactuated, contacts are hybrid and nonsmooth, and a one-step reference increment may be physically infeasible from an off-reference state. A generic MLP can freely mix $\mathbf{e}_t$ and $\mathbf{m}_t$; the terms are feedback- and feedforward-aligned coordinates, not identifiable controller branches. Rotation differences are chords in the chosen embedding, not general Lie-algebra tangents. The policy remains reference conditioned and does not define a reference-blind autonomous skill. Finally, evidence from single-motion or single-seed studies cannot establish a general optimization advantage; the confirmatory suite and strict paired protocol are essential.

The equivalence theorem is conditional on access to $\mathbf{x}_t$ and on invertible preprocessing. It does not apply to an official target interface that omits velocity features or uses a different lookahead. Nor does it imply that every feature coordinate is controllable in one step: root motion is underactuated and contact impulses can dominate the immediate response. Our completion definitions are external to the learned reward, but remain task-specific evaluation choices and must be fixed before inspecting method outcomes. Finally, the present study uses one simulated humanoid and motion-capture references. Transfer to different morphologies, noisy state estimation, actuation delay, and hardware requires separate validation.

C. Implications for Robot Learning

Many robot-learning systems specify goals in one representation but evaluate behavior in another. The present analysis suggests a simple design question: can the command be expressed in coordinates that close under the task residual without adding information? This principle may apply to end-effector tracking, legged-locomotion commands, visual servoing, and model-predictive residuals. Those extensions require task-specific feature charts and remain future work.

D. Computational Cost and Reproducibility

The command construction uses only feature subtraction and fixed diagonal scaling. It adds no learned parameters and requires no additional simulator steps. We nevertheless report environment throughput, policy latency, and peak memory for both representations. The release will include the feature ordering, frozen scales, inverse-transform unit test, paired seed list, final-checkpoint evaluator, and scripts that regenerate every table and curve. **[PLACEHOLDER—NOT A RESULT: Insert measured throughput, latency, memory, and artifact URL.]**

IX. CONCLUSION

We study the coordinate interface between a reference-conditioned policy and a learned differential tracking reward. Absolute and residual one-step commands contain exactly the same information and support the same policy class, but the

residual command obeys an exact post-action closure identity with the reward. This yields a minimal intervention for testing how representation shapes finite policy optimization. **[PLACEHOLDER—NOT A RESULT: Replace with the final quantitative conclusion across motions and seeds, including any negative or mixed result.]**

ACKNOWLEDGMENT AND AI-USE DISCLOSURE

This anonymous draft contains no identifying acknowledgment. A generative AI assistant was used to help structure and edit prose and LaTeX; all scientific claims, equations, citations, and experimental results are to be verified and approved by the authors. This disclosure will be updated to match the final ICRA policy and actual use.

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